MICROSTRIP PATCH ANTENNA FOR BLUETOOTH/ZIGBEE IoT NODES

Fabricated on FR-4 PCB — 2.45 GHz Inset-Fed Patch Antenna: Design, Simulation & Fabrication-Ready Validation in ANSYS HFSS

Team members:

Khanna M (24BEC0187)
Nishanth A (24BEC0595)
Sashank P S (24BEC0636)
Tarun Sathish (24BEC0631)
Shailesh A (24BEC0665)

Group Project • ANSYS Electronics Desktop 2025 R2.4 (Student)

0

OVERVIEW

Agenda

1 **Introduction & Motivation**
Why low-cost FR-4 patch antennas suit IoT nodes

2 **Project Objectives**
What the design and simulation set out to achieve

3 **Design Parameters**
Substrate, patch, and feed geometry

4 **Simulation Setup**
HFSS modeling and solve methodology

5 **Results**
Return loss, VSWR, and radiation performance

6 **Performance Summary**
All metrics vs. requirements at a glance

7 **Conclusion & Fabrication Readiness**
What's proven, and how it moves to a PCB prototype

8 **Future Work & Team**
Next steps, and who did what

1

BACKGROUND

Introduction & Motivation

- IoT sensor nodes for Bluetooth Low Energy (BLE) and Zigbee (IEEE 802.15.4) applications need antennas that are compact, low-cost, and easy to mass-produce alongside the rest of the PCB assembly.
- A microstrip patch antenna printed directly onto standard FR-4 substrate meets all three needs: it uses the same low-cost material and fabrication process as the host PCB, requires no separate antenna component, and keeps the whole design low-profile and planar.
- The 2.4–2.4835 GHz ISM band is shared by BLE and Zigbee, so a single antenna tuned to 2.45 GHz can serve both radio standards — a common choice for smart-home, industrial, and wearable IoT nodes.
- This project designs, simulates, and validates such an antenna in ANSYS HFSS, with a layout intended for direct fabrication on 1.6 mm FR-4 PCB using a standard 50 Ω SMA edge connector.

WHY A PATCH ON FR-4?

Low cost

FR-4 is the standard, cheapest PCB substrate available.

Planar & compact

Etched directly onto the PCB — no separate antenna part.

Dual-standard

2.45 GHz covers both BLE and Zigbee in one design.

Manufacturable

Standard photolithography / PCB milling, SMA connector.

2

SCOPE

Project Objectives

01

Design

Design a compact 2.45 GHz microstrip patch antenna on low-cost FR-4 substrate, matched to a 50 Ω feed.

02

Simulate

Model and solve the structure in ANSYS HFSS using Driven Modal analysis across the 2.0–3.0 GHz band.

03

Validate

Confirm resonance, impedance match (VSWR), and radiation behavior meet ISM-band requirements.

04

Prepare for Fabrication

Finalize a parametric, manufacturable layout ready to be etched on FR-4 PCB with an SMA feed.

3

DESIGN

Design Parameters & Material Specification

Parameter	Symbol	Value	Unit	Purpose
Operating Frequency	f_o	2.45	GHz	Bluetooth 5.0 / BLE / Zigbee ISM band
Substrate Material	—	FR-4 Epoxy	—	Standard low-cost PCB material
Dielectric Constant	εr	4.4	—	Relative permittivity of FR-4
Loss Tangent	tan δ	0.02	—	Dielectric loss factor
Substrate Thickness	h	1.60	mm	Standard 1.6 mm PCB height
Substrate Dimensions	SubX × SubY	60.0 × 60.0	mm	Overall ground plane area
Patch Width	W	37.41	mm	Controls input resistance
Patch Length	L	28.85	mm	Controls 2.45 GHz resonant frequency
Feed Line Width	Wf	3.02	mm	Gives 50 Ω characteristic impedance
Inset Depth	Fi	9.20	mm	Impedance-matching feed insert depth
Notch Gap	g	1.00	mm	Inset cutout clearance gap
Port Type	—	Lumped Port	—	50 Ω matched excitation (SMA at fabrication)

4

MODEL

3D Antenna Geometry & Parametric Setup

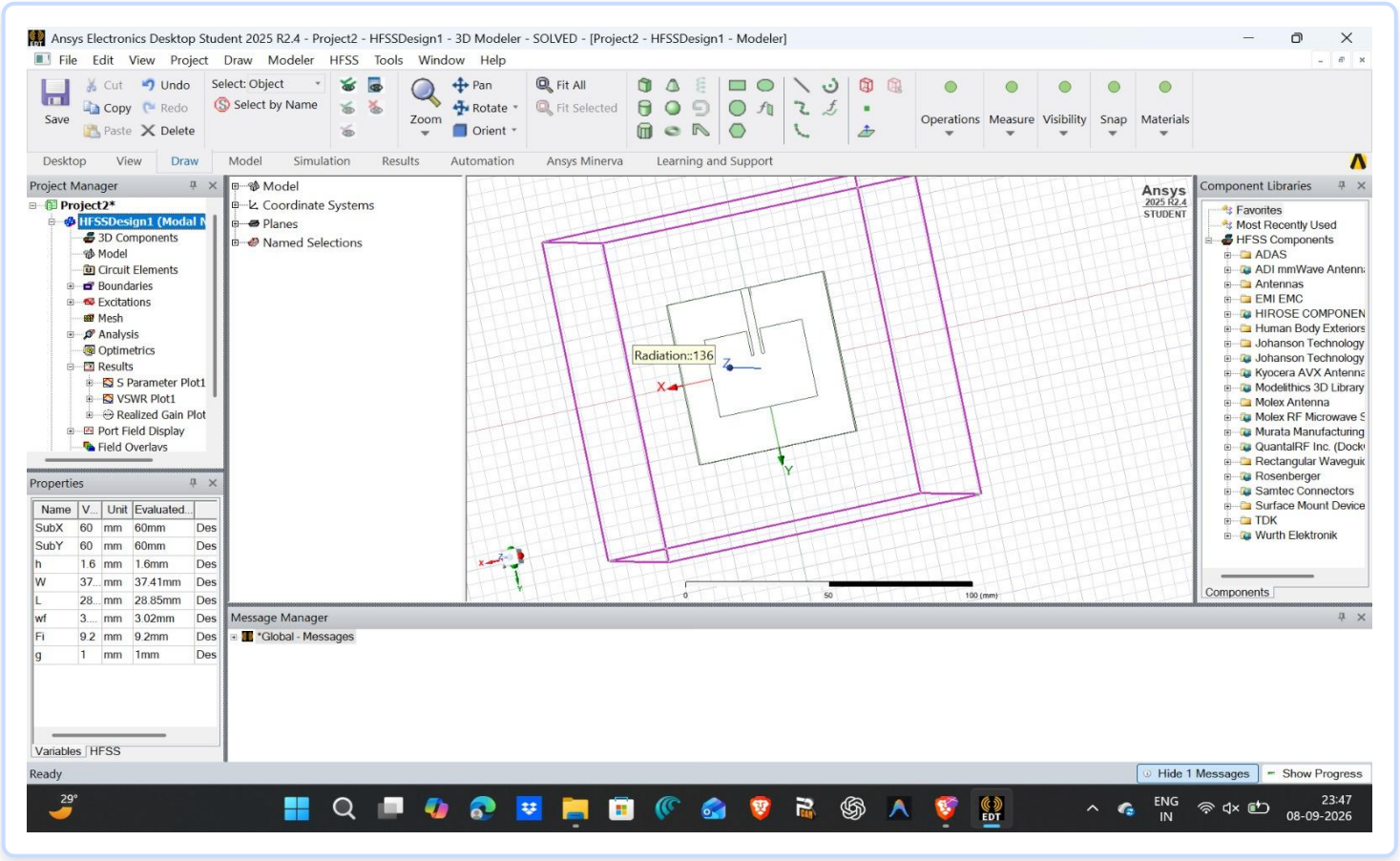


Figure 1: HFSS 3D model of the inset-fed patch antenna, with the parametric variable table and radiation boundary shown.

Substrate

60 × 60 × 1.6 mm FR-4

Patch

37.41 × 28.85 mm

Feed line

3.02 mm wide, 9.20 mm inset

Air box

Radiation boundary for open-space simulation

5

METHOD

Simulation Methodology

An inset-fed microstrip patch antenna operating at 2.45 GHz was modeled and simulated in ANSYS HFSS (Electronics Desktop, Student 2025 R2.4).

1 Substrate & Ground

Double-sided FR-4 epoxy substrate (εr = 4.4, h = 1.6 mm) with a full ground plane on the bottom layer.

2 Parametric Modeling

Design variables (SubX, SubY, W, L, Wf, Fi, g) were used throughout, enabling fast optimization sweeps.

3 Impedance Matching

An inset feed (depth Fi = 9.20 mm, gap g = 1.00 mm) matches the 50 Ω microstrip line to the patch.

4 Excitation

A 50 Ω lumped port was assigned across a vertical integration line from ground plane to feed line.

5 Radiation Boundary

An air box surrounding the structure was assigned a radiation boundary to emulate open-space propagation.

6 Solve Setup

Driven Modal solution, discrete frequency sweep 2.0–3.0 GHz, 5 MHz (0.005 GHz) step resolution.

6

RESULTS

Return Loss (S_{11}) Performance

-16.40 dB

S_{11} at resonance (marker m1)

2.43 GHz

Simulated resonant frequency

2.400–2.4835 GHz

−10 dB impedance bandwidth

What the S_{11} sweep shows

- A sharp resonance dip of −16.40 dB occurs at 2.43 GHz, well below the standard −10 dB pass threshold, confirming a strong impedance match at resonance.
- The −10 dB bandwidth spans 2.400–2.4835 GHz — fully covering the ISM band required for Bluetooth Low Energy (BLE) and Zigbee (IEEE 802.15.4), despite the resonance sitting 20 MHz below the 2.45 GHz design target.
- This small frequency shift is typical of first-pass simulations and is easily corrected by trimming the patch length (L) slightly before fabrication — shorter L raises the resonant frequency.

7

RESULTS

Impedance Matching (VSWR)

1.36
VSWR at 2.43 GHz (marker m1)

< 1.5
Target VSWR threshold

> 95%
Power delivered into antenna

Reading the VSWR curve

- VSWR falls to 1.36 at 2.43 GHz, comfortably under the VSWR ≤ 1.5 criterion generally used to define a well-matched antenna.
- A VSWR this low means over 95% of the incident power from the 50 Ω feed is transferred into the antenna, with minimal reflection back toward the source — important for battery-powered IoT nodes where every milliwatt matters.
- Away from resonance, VSWR rises sharply toward the edges of the 2.0–3.0 GHz sweep (>25), which is expected — the patch is a narrowband resonant structure by design.

8

RESULTS

Radiation Pattern & Simulation Validation

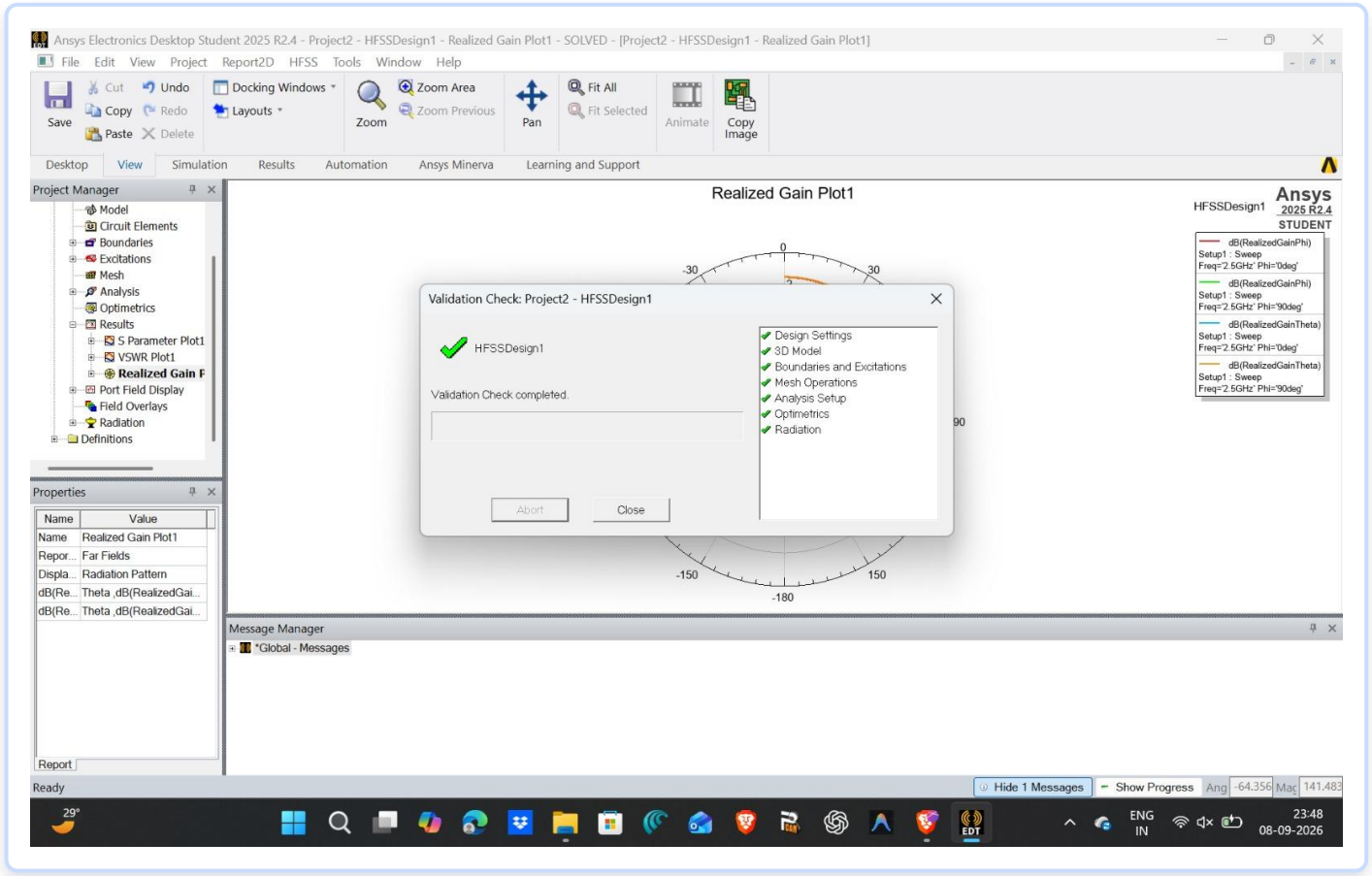


Figure 2: HFSS validation check (all categories passed), with the 2.45 GHz realized-gain polar pattern behind it.

Validation Checklist

- ✓ Design Settings
- ✓ 3D Model
- ✓ Boundaries & Excitations
- ✓ Mesh Operations
- ✓ Analysis Setup
- ✓ Optimetrics
- ✓ Radiation

Radiation behavior

Broadside pattern with peak realized gain directed at θ = 0° (perpendicular to the patch). Low cross-polarization and back-lobe levels indicate high radiation purity — suitable for IoT sensor nodes.

9

RESULTS

Performance Summary

Metric	Simulated Result	Requirement	Status
Resonant Frequency	2.43 GHz (target 2.45 GHz)	2.4–2.4835 GHz ISM band	**Pass**
Return Loss (S_{11})	−16.40 dB	≤ −10 dB	**Pass**
−10 dB Bandwidth	2.400–2.4835 GHz (~84 MHz)	Covers full ISM band	**Pass**
VSWR	1.36	≤ 1.5	**Pass**
Radiation Pattern	Broadside, low back-lobe	Peak at θ = 0°	**Pass**
HFSS Validation Check	7 / 7 categories passed	All categories green	**Pass**

10

CONCLUSION

A Fabrication-Ready 2.45 GHz Antenna for IoT Nodes

An inset-fed microstrip patch antenna was successfully designed, simulated, and analyzed in ANSYS HFSS for Bluetooth- and Zigbee-class IoT applications. The antenna resonates at 2.43 GHz with a VSWR of 1.36 and strong broadband, broadside radiation. The parametric layout is fabrication-ready for a standard double-sided 1.6 mm FR-4 PCB with a 50 Ω SMA connector, making it suitable for direct integration into compact IoT sensor nodes.

- 2.43 GHz Resonance
- −16.40 dB Return Loss
- VSWR 1.36
- Broadside Radiation
- 7/7 Validation Checks

With the frequency offset trimmed and the layout etched onto FR-4, this design is ready to move from simulation to a physical, measurable prototype (see Future Work).

11

ROADMAP

Future Work & Next Steps

1

Fine-tune resonance

Trim patch length (L) by a fraction of a mm to pull resonance from 2.43 GHz to exactly 2.45 GHz.

2

Fabricate the PCB

Etch the validated layout onto 1.6 mm FR-4 using standard photolithography or PCB milling.

3

Mount the SMA connector

Solder an edge-launch 50 Ω SMA connector at the feed point for lab measurement.

4

Measure on a VNA

Measure S_{11} and VSWR on a vector network analyzer and compare directly against the HFSS results.

5

Verify the radiation pattern

Test the fabricated prototype in an anechoic chamber or open-field range against the simulated pattern.

6

Integrate & field-test

Connect to a BLE/Zigbee module and test real-world link range and RSSI inside a housed IoT node.

Thank You

Questions & Discussion